

Outline (until lunch break)

- Hands-on: Association of Barcode/Tag Sequences
- Hands-on: Count Sequencing Analysis
- Discussion of QC metrics
- Hands-on: Data Analysis Steps (Regions and Variants)

Hands-on: Association of Barcode/Tag Sequences



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Max Schubach
Computational Genome Biology @ BIH

July 20, 2025



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Running the association workflow



03 Standard Pipeline for MPRA Analysis: MPRAsnakeflow

External links:

- [MPRAsnakeflow repository](#)
- [MPRAsnakeflow documentation](#)
- [MPRAsnakeflow tutorial](#)

Hands-on:

- Association of Barcode/Tag Sequences: [Jupyter/Colab notebook](#)
- Count Sequencing Analysis: [Jupyter/Colab notebook](#)



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Thank You! Questions?

kircherlab.github.io

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Hands-on: Count Sequencing Analysis



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Discussion of QC metrics

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Discussion of QC metrics

1. QC report assignment workflow:

- <https://kircherlab.github.io/mprasnakeflow/assignment.html>

2. QC report experiment workflow:

- <https://kircherlab.github.io/mprasnakeflow/experiment.html>

MPRAsnakeflow assignment QC report

Code Table of contents

Overview

The Assignment QC (Quality Control) Report is a comprehensive document designed to evaluate and ensure the quality of data or processes involved in a specific assignment. This report is generated for the assocBasic assignment with settings defined in the config default. The barcode length for this experiment is 15 bases.

- **Forward:** data/SRR10800986_1.fastq.gz
- **Reverse:** data/SRR10800986_3.fastq.gz
- **Barcode:** data/SRR10800986_2.fastq.gz
- **Design File:** design.fa

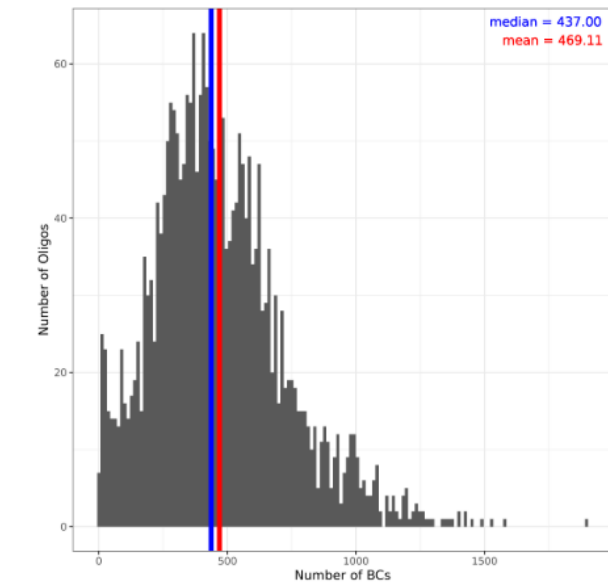
Overall quality metrics

Table explanation

- **median assigned barcodes:** Median number of barcodes assigned to tested sequences in mapping as a quality control measure for the assignment step, whether there is sufficient barcode to oligo coverage.
- **fraction assigned oligos:** Fraction of assigned tested sequences in mapping to determine if the library during the assignment step was sufficiently recovered.

median assigned barcodes	fraction assigned oligos
437	1.00

BC distribution over oligos



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MPRA analysis with BCalm

ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

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Hands-on: Identifying regions and variants with significant effects



04 MPRA Data Analysis

Hands-on:

- Data Analysis Steps (Regions and Variants): [README](#), [Jupyter/Colab notebook](#)

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

What are we looking for?

Element Analysis: *"Is this sequence regulatory active?"*
Variant Analysis: *"Does this variant change regulatory activity?"*

Element 1	...ATCGC...		...ATCGC...		Reference
Element 2	...CGTCC...		...AT T GC...		Variant 1
Element 3	...CAGCT...		...AT A GC...		Variant 2
Neg Control	...TGCAA...		...AT G GC...		Variant 3
Expression			Expression		

Revisiting the data structure from MPRA experiment

Barcode level:

- Each test element represented by many barcodes
- High variance of the RNA/DNA ratio per barcode
- Barcode sequence might influence activity

Oligo level:

- Robust activity measurement for oligos
- Outlier barcodes may indicate sequence-specific effects?

Replicate level:

- Technical replicates for experimental reproducibility
- Replicate sample normalized for read depth

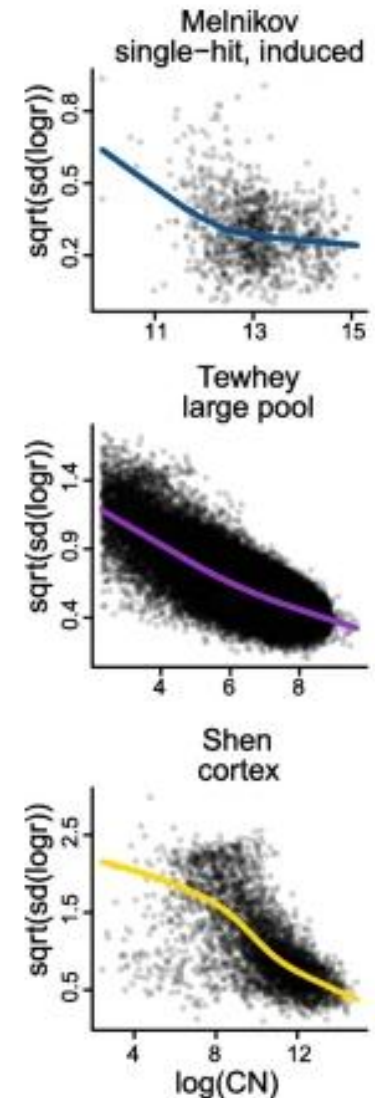
Barcode	name	dna_count_1	rna_count_1
AAAAAAAAAAGCTGC	oligo_006364	11	10
AAAAAAAAATCACAG	oligo_005641	8	7
AAAAAAAAATTGAGC	oligo_005719	2	25
AAAAAAAAATTGTAT	oligo_005725	4	12
AAAAAAAACACATGA	oligo_005412	21	15
AAAAAAAACATAGTT	oligo_006471	8	3

Revisiting the data structure from MPRA experiment

Counts follow a distribution where the variance increases with the mean

Elements with lower DNA counts have much more variable activity measurements

Different approaches have different statistical properties - average estimator OR aggregate estimator?



mpralm - linear models to analyze MPRA data

General tools repurposed for MPRA analysis

- DESeq2 for significant transcriptional activity
- t-test for significant differential activity

MPRA analysis specific tools

- MPRAnalyze
- mpralm
- BCalm

But why?

voom

mpralm

empirical Bayes

Method | [Open access](#) | Published: 02 September 2019

MPRAnalyze: statistical framework for massively parallel reporter assays

[Tal Ashuach](#), [David S. Fischer](#), [Anat Kreimer](#), [Nadav Ahituv](#), [Fabian J. Theis](#) & [Nir Yosef](#) 

[Genome Biology](#) **20**, Article number: 183 (2019) | [Cite this article](#)

Methodology article | [Open access](#) | Published: 12 March 2019

Linear models enable powerful differential activity analysis in massively parallel reporter assays

[Leslie Myint](#), [Dimitrios G. Avramopoulos](#), [Loyal A. Goff](#) & [Kasper D. Hansen](#) 

[BMC Genomics](#) **20**, Article number: 209 (2019) | [Cite this article](#)

Research | [Open access](#) | Published: 13 February 2025

Using individual barcodes to increase quantification power of massively parallel reporter assays

[Pia Keukeleire](#), [Jonathan D. Rosen](#), [Angelina Göbel-Knapp](#), [Kilian Salomon](#), [Max Schubach](#) & [Martin Kircher](#) 

BCalm - *mpralm*, but more robust against outliers

Problem with aggregation:

- When you sum up barcode counts, you lose information about variability between barcodes
- A single outlier barcode can heavily influence the aggregated result

BCalm's solution:

- Keep each barcode as a separate observation in the statistical model
- Gain statistical power from having more data points
- Robust handling of outlier barcodes through the modeling framework

Barcode	name	dna_count_1	rna_count_1
AAAAAAAAAAGCTGC	oligo_006364	11	10
AAAAAAAAATCACAG	oligo_005641	8	7
AAAAAAAAATTGAGC	oligo_005719	2	25
AAAAAAAAATTGTAT	oligo_005725	4	12
AAAAAAAACACATGA	oligo_005412	21	15
AAAAAAAACATAGTT	oligo_006471	8	3

And now we will ...

- Explore the dataset in detail
- Perform element analysis using BCalm
- Perform variant analysis using BCalm

Thank You!

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